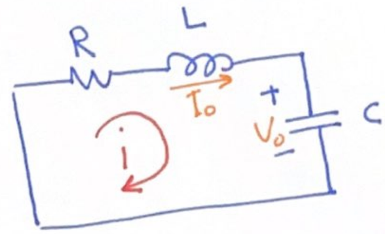
The Source_Free Series RLC

I_o, V_o are the initial inductor current and the initial capacitor voltage that excite the circuit.



At time $t = 0$

$$i\Big|_{t=0} = I_o \quad , \quad V\Big|_{t=0} = \frac{1}{C}\int_{-\infty}^{t=0} i\,d\tau = V_o$$

Apply KVL $\quad V_R + V_L + V_C = 0$

$V_R = iR$

$V_L = L\frac{di}{dt}$

$V_C = \frac{1}{C}\int i\,dt$

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt \; + V_C(0) = 0$$ Differentiate

$$R\frac{di}{dt} + L\frac{d^2 i}{dt} + \frac{1}{C} i = 0$$ Re arrane

$$\frac{d^2 i}{dt} + \frac{R}{L}\frac{di}{dt} + \frac{1}{LC} i = 0$$ Second_order D.E. (Linear, homogeneous)

Assume the solution is of form $i = e^{st}$

→ As one experience from the first-order.

→ to get the benift of its derivatives

$\frac{di}{dt} = s e^{st}$

$\underbrace{\frac{d^2 i}{dt} = s^2 e^{st}}_{\text{multiput of the function}}$

$$s^2 e^{st} + \frac{R}{L} s e^{st} + \frac{1}{LC} e^{st} = 0$$

$$e^{st} \underbrace{\left(s^2 + \frac{R}{L} s + \frac{1}{LC}\right)}_{\text{quadratic equation.}} = 0$$

Only the quadratic equation can make the left hand side equal to zero for all t.

Using quadratic formula $ax^2+bx+c=0$

$$x_1, x_2 = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$s_1, s_2 = \frac{-\frac{R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}}}{2} = \frac{-R}{2L} \pm \frac{\sqrt{\left(\frac{2R}{2L}\right)^2 - \frac{4}{LC}}}{2}$$

$$s_1, s_2 = \frac{-R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \left(\frac{1}{\sqrt{LC}}\right)^2}$$

Let $\boxed{\alpha = \frac{R}{2L} \qquad w_0 = \frac{1}{\sqrt{LC}}}$

$$\boxed{\begin{aligned} s_1 &= -\alpha + \sqrt{\alpha^2 - w_0^2} \\ s_2 &= -\alpha - \sqrt{\alpha^2 - w_0^2} \end{aligned}}$$ (general Roots)

s_1, s_2 : natural frequencies / nepers per second N_p/s

w_0 : resonant frequency / radians per second rad/s
Undamped natural frequency

α : neper frequency / Nepers Per second N_p/s
damping factor.

The quadratic equation can be re-arrang to:

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0 \equiv s^2 + 2\alpha s + w_0^2 = 0$$

Now because we have two roots for the quadratic equation, the we could have two possible solutions.

$$i_1 = e^{s_1 t} \quad , \quad i_2 = e^{s_2 t}$$

i_1 and i_2 are two linearly independent solutions of a Linear, homogeneous second order D.E., then the general solution is

$$\boxed{i_{(t)} = A_1 e^{s_1 t} + A_2 e^{s_2 t}}$$

For two Real Roots

A_1, A_2 arbitrary constants determind from the initial valus- $i\big|_{t=0}$, $\frac{di}{dt}\big|_{t=0}$

From the initial values

$$i\big|_{t=0} = I_o \quad \text{-----} \quad (1)$$

$$R i_{(0)} + L \frac{di_{(0)}}{dt} + \underbrace{\frac{1}{C}\int i_{(0)}\, dt}_{V_{(0)}} \Rightarrow L \frac{di_{(0)}}{dt} = -R I_o - V_{(0)}$$

$$\frac{di}{dt}\bigg|_{t=0} = -\frac{1}{L}\left(R I_o + V_o\right) \quad \text{-----} \quad (2)$$

From the general Roots, we can notice that there are three Solutions

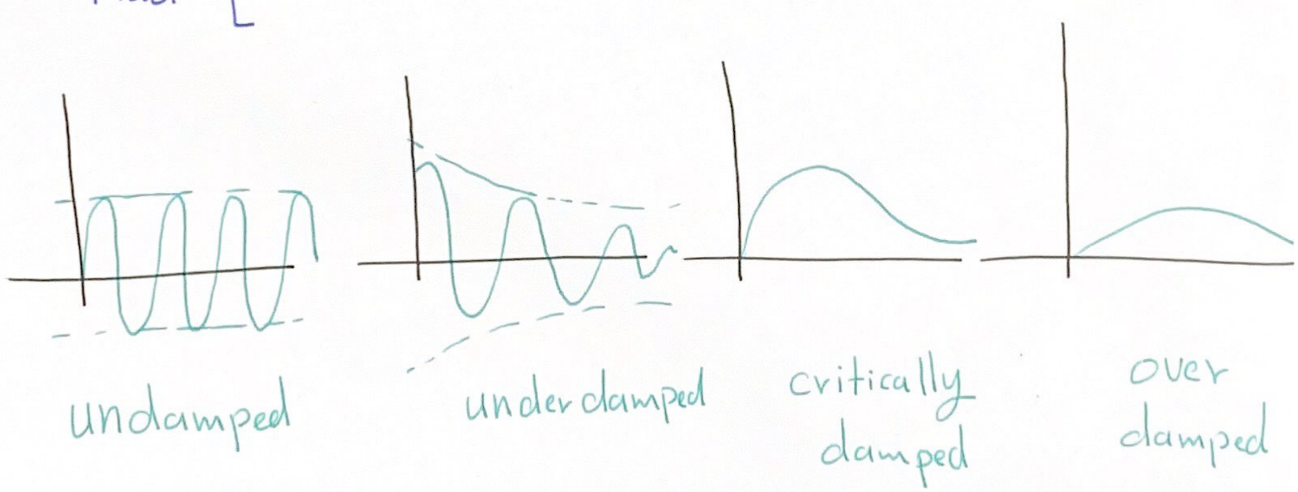
1- $\alpha > \omega_o$ (overdemped case)

2- $\alpha = \omega_o$ (critically damped case)

3- $\alpha < \omega_o$ (underdamped case)

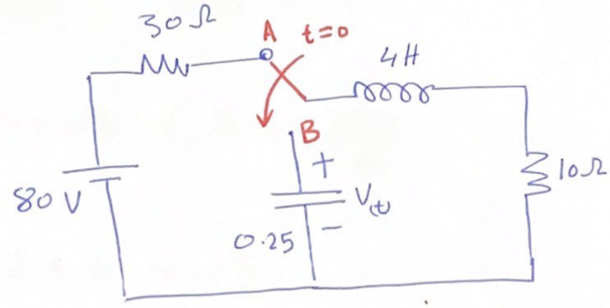
Damping :- The gradual loss of the initial stored energy, as noteced by the continuous decrease in the amplitude of the response.

- Damping is due to the presence of resistance R
- The damping factor α determines the rate at which the response is damped. $\alpha = \frac{R}{2L}$
- IF $R=0$, then $\alpha=0$, $\Rightarrow$ LC circuit (loss-less). The response is undamped and oscillatory. (infinite oscillation).
- By adjusting the value of R, the response can be mad [

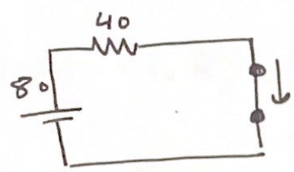


Ex: The switch moves from Position A to Position B at $t=0$

Let $V_{(0)}=0$, find $V_{(t)}$ for $t>0$.



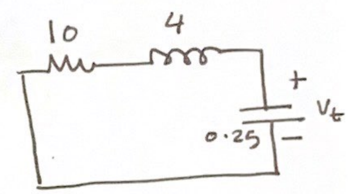
In Position A we have RL circuit in Steady state mode



$$i_L = \frac{V}{R} = \frac{80}{40} = 2A$$ and as $i_{(0)}^- = i_0$ this current will be initial current at B.

In Position B

We have a source Free series RLC circuit.



$$S_{1,2} = -\alpha \pm \sqrt{\alpha^2 - w_0^2} \quad Np/s$$

$$\alpha = \frac{R}{2L} = \frac{10}{2*4} = 1.25 \quad Np/s$$

$$w_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{1}{4}\cdot 4}} = 1 \quad rad/s$$

$$S_{1,2} = -1.25 \pm \sqrt{(1.25)^2 - 1^2} = -1.25 \pm 0.75$$

$S_1 = -2$

$S_2 = -0.5$

As two Real Roots, the general solution is

$$i = A e^{-2t} + B e^{-0.5t}$$

(Next we have to find A and B)

We use the initial conditions to find A and B

$i_o = 2\,A \qquad V_{(0)c} = 0 \qquad$ and $\qquad V_R + V_L + V_c = 0$

$iR + L\frac{di}{dt} + V_c = 0$ at $t=0 \Rightarrow i_o R + L\frac{di_o}{dt} + \underbrace{V_{C(0)}}_{=0} = 0$

$$\frac{di_o}{dt} = \frac{-1}{L}(i_o R) = \frac{-1}{4} * 2 * 10 = -5$$

Now Back to our current equation.

$$i = A e^{-2t} + B e^{-0.5t}$$

① at $t=0 \quad i\Big|_{t=0} = 2$

$$\Rightarrow \quad i_o = A + B = 2 \quad ---- ①$$

② $\frac{di}{dt} = -2Ae^{-2t} - \frac{B}{2}e^{-0.5t}$ and $\frac{di_o}{dt} = -5$

$$-2A - \frac{B}{2} = -5 \quad ---- ②$$

from 1 and 2 $\qquad \begin{matrix} A + B = 2 \\ -2A - \frac{1}{2}B = -5 \end{matrix} \Rightarrow \begin{matrix} B = -0.666 \\ A = 2.666 \end{matrix}$

$$i_{(t)} = 2.666\,e^{-2t} - 0.666\,e^{-0.5t}$$

$V_R + V_L + V_c = 0$ and $V_c = V_t = V_R + V_L$

$= iR + L\frac{di}{dt}$

$R = 10\Omega$

$$iR = 26.66\,e^{-2t} - 6.66\,e^{-0.5t}$$

$$L\frac{di}{dt} = 4\left[-5.332\, e^{-2t} + 0.333\, e^{-0.5t}\right]$$

$$= -21.332\, e^{-2t} + 1.332\, e^{-0.5t}$$

$$V_t = iR + L\frac{di}{dt} = 5.328\, e^{-2t} - 5.32\, e^{0.5t}$$

OR

$$V_t = A e^{-2t} + B e^{-0.5t}$$

$V_0 = 0 \Rightarrow A + B = 0$ ---- (1)

$i_c = \frac{c\,dv}{dt} \Rightarrow \frac{dv}{dt} = \frac{i_c}{c}$ and $\frac{dv_0}{dt} = \frac{i_0}{c} = \frac{-2}{0.25}$

$\frac{dv_0}{dt} = -8$

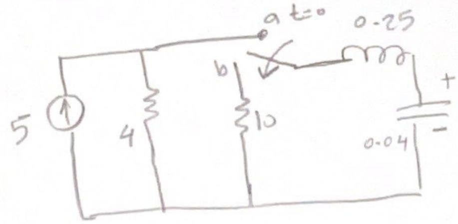
$\frac{dV}{dt} = -2A e^{-2t} - \frac{1}{2} B e^{-0.5t}$ and $-2A e^{-2t} - \frac{1}{2} B e^{-0.5t}$

$-2A - \frac{1}{2}B = -8$ --- (2) From 1 and 2

$A = 5.33$
$B = -5.33$

$$V_t = 5.33\, e^{-2t} - 5.33\, e^{-0.5t}$$

EX: Find $V_{(t)}$ For $t \geq 0$



$I_0 = 0$, $V_0 = 20$

$i_{(t)} = 2.309\,(e^{-37.32t} - e^{-2.67t})$

$V_t = 21.55\, e^{-2.67t} - 1.55\, e^{-37.3t}$